Project Report - Day 8

Title : **Architecting a Retrieval-Augmented Generation (RAG) Pipeline**

Objective

Build a RAG pipeline where a user query is answered strictly from a provided document.

- Load the document (text / optionally PDF).
- Get embeddings for chunks via SentenceTransformer.
- Store chunks + embeddings in vector store (ChromaDB).
- At query time, embed query and retrieve nearest chunk(s).
- Send chunk(s) + query to LLM prompt with explicit grounding instruction:
    - "Answer using ONLY provided context. If answer not in context, say I DON'T KNOW."
- If LLM unavailable or fails, fallback to strict "I DON'T KNOW." Behavior.

Libraries Used

1. **pathlib**
    - Used to build a reliable file path for loading document.txt from the same folder as the script.
2. **chromadb**
    - Vector store for chunks embeddings and similarity search.
3. **os**
    - Used to handle file paths and access the .env file location.
4. **Sentence-transformers**
    - All-MiniLM-L6-V2 to convert text/corpus/query into embeddings.
5. **dotenv**
    - Used to load environment variables from the .env file.

Errors

1. **Gemini model not found / 404**
    - **Issue:** request to invalid model (gemini-1.5-flash in v1beta).
    - **Solution:** fallback to local context-only behavior. Also use supported model name (e.g., gemini-2.5-flash or as available).
2. **document.txt not found**
    - **Issue:** running script from wrong working dir.
    - **Solution:** set load_document() path to Path(__file__).parent / "document.txt" and run in Day8_Task..
3. **No context or false "I DON'T KNOW" due retrieval mismatch**
    - **Issue:** n_results() or chunk cut into wrong unnits.
    - **Solution:** increase n_units and ensure chunk sizes (100-200 words) and accurate query embedding retrival.
4. **Chroma collection initialization errors**
    - **Issue:** collection exists and query returns empty or old state.
    - **Solution:** drope recreate collection each run with client.delete_collection("rag_collection") safe try/except.

OUTPUT

```
PS D:\Daily_Tasks> & C:\Users\Acer\AppData\Local\Programs\Python\Python310\python.exe d:/Daily_Tasks/Day8_Task/rag.py

You: What is this document about?

Answer:
This document is about the NexusVortex Operating System, its Pulse-Core synchronization layer, temporal sharding, and the Aegis Security
Layer. It details the responsibilities of specific shards and the partitioning requirements for the primary drive.

Query: What is this document about?

Query Embedding:
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You: exit
PS D:\Daily_Tasks>
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